Rozwiązania wybranych przykładów z Z08

3. Oblicz całki nieoznaczone (dodawanie, mnożenie przez stałą)

$$\int x^p dx = \frac{1}{p+1}\cdot x^{p+1} + C, \quad \int dx = x + C\,; \quad \int x\,dx = \frac{1}{2}x^2 + C$$
$$\int \frac{1}{x} dx = \ln|x| + C$$
$$\int e^x dx = e^x + C, \qquad \int a^x dx = a^x / \ln a + C$$
$$\int \sin x dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C$$

$a)$ $\int \frac{1}{e^x} dx = \int e^{-x} dx = -e^{-x} + C$

$b)$ $\int \frac{x^4 - x^3 + 2x}{x} dx = \int (x^3 - x^2 + 2)\,dx = \frac{1}{4}x^4 - \frac{1}{3}x^3 + 2x + C$

$c)$ $\int \frac{x + 2\sqrt[3]{x} - \sqrt{x^3}}{\sqrt{x}} dx = \int \frac{x}{\sqrt{x}} dx + 2\int \frac{\sqrt[3]{x}}{\sqrt{x}} dx - \int \frac{\sqrt{x^3}}{\sqrt{x}} dx =$

$= \int x^{\frac{1}{2}} dx + 2\int x^{-\frac{1}{6}} dx - \int x\,dx = \frac{2}{3}x^{\frac{3}{2}} + \frac{12}{5}x^{\frac{5}{6}} - \frac{1}{2}x^2 + C$

$e)$ $\int \frac{(x - x^2)^2}{x^2} dx = \int \frac{x^2 - 2x^3 + x^4}{x^2} dx = \int (1 - 2x + x^2)\,dx = \cdots$

$f)$ $\int 3^x\,dx = \frac{3^x}{\ln 3} + C$

$g)$ $\int \frac{2^x - 3^x}{10^x} dx = \int \left(\frac{2}{10}\right)^x dx - \int \left(\frac{3}{10}\right)^x dx = \frac{\left(\frac{1}{5}\right)^x}{\ln\left(\frac{1}{5}\right)} - \frac{\left(\frac{3}{10}\right)^x}{\ln\left(\frac{3}{10}\right)} + C$

$i)$ $\int \frac{1}{1 - x^2}\,dx = |\textit{rozkładamy ułamek na sumę}\,| = \frac{1}{2}\int \frac{1}{1-x} + \frac{1}{1+x}\,dx =$

$$\frac{1}{2}(-\ln|1 - x| + \ln|1 + x|) + C.$$

4. Oblicz całkując przez części:

Całkowanie przez części

$$\int f \cdot g'dx = \begin{vmatrix} f = \cdots & g' = \cdots \\ f' = \cdots & g = \cdots \end{vmatrix} = f \cdot g - \int f' \cdot g\,dx = \ldots$$

Przykład:

$$\int x\cos x\,dx = \begin{vmatrix} f = x & g' = \cos x \\ f' = 1 & g = \sin x \end{vmatrix} = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$$

$$\int x\cos x\,dx = \begin{vmatrix} f = x & g' = \cos x \\ f' = 1 & g = \sin x \end{vmatrix} = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C.$$

$$c)\ \int x^2\sin x\,dx = \begin{vmatrix} f = x^2 & g' = \sin x \\ f' = 2x & g = -\cos x \end{vmatrix} = -x^2\cos x + \int 2x\cos x\,dx =$$

$$= \begin{vmatrix} f = 2x & g' = \cos x \\ f' = 2 & g = \sin x \end{vmatrix} = -x^2\cos x + 2x\sin x + 2\cos x + C.$$

$$d)\ \int xe^{-2x}\,dx = \begin{vmatrix} f = x & g' = e^{-2x} \\ f' = 1 & g = -\frac{1}{2}e^{-2x} \end{vmatrix} = -\frac{1}{2}xe^{-2x} + \frac{1}{2}\int e^{-2x}dx =$$

$$= -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + C$$

$$e)\int \ln x\,dx = \int 1\cdot\ln x\,dx = \begin{vmatrix} f = \ln x & g' = 1 \\ f' = \frac{1}{x} & g = x \end{vmatrix} = x\ln x - \int x\cdot\frac{1}{x}dx = x\ln x - \int 1\,dx$$

$$= x\ln x - x + C = x(\ln x - 1) + C$$

$$j)\ \int \frac{\ln x}{x^2}dx = \begin{vmatrix} f = \ln x & g' = \frac{1}{x^2} \\ f' = \frac{1}{x} & g = -\frac{1}{x} \end{vmatrix} = -\frac{\ln x}{x} + \int \frac{1}{x^2}dx = -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$l^*)\ \int \ln(x^2+1)\,dx = \begin{vmatrix} f = \ln(x^2+1) & g' = 1 \\ f' = \frac{2x}{x^2+1} & g = x \end{vmatrix} = x\ln(x^2+1) - \int x\frac{2x}{x^2+1}dx =$$

$$= x\ln(x^2+1) - 2\int\left(1 - \frac{1}{x^2+1}\right)dx = x\ln(x^2+1) - 2x + 2\operatorname{arctg} x + C$$

$$m)\ \int (\ln x)^2\,dx = \begin{vmatrix} f = \ln^2 x & g' = 1 \\ f' = 2\ln x\cdot\frac{1}{x} & g = x \end{vmatrix} = x\ln^2 x - 2\int \ln x\cdot\frac{1}{x}xdx =$$

$$= \begin{vmatrix} f = \ln x & g' = 1 \\ f' = \frac{1}{x} & g = x \end{vmatrix} = x\ln^2 x - 2\left(x\ln x - \int x\frac{1}{x}dx\right) = x\ln^2 x - 2x\ln x + 2x + C$$

$$n^*)\int \sqrt{x}\cdot\cos(\sqrt{x})\ dx = \int 2x\cdot\frac{\cos(\sqrt{x})}{2\sqrt{x}}dx = \begin{vmatrix} f = 2x & g' = \frac{\cos\sqrt{x}}{2\sqrt{x}} \\ f' = 2 & g = \sin(\sqrt{x}) \end{vmatrix} =$$

$$= 2x\sin\sqrt{x} - 2\int \sin\sqrt{x}\ dx = \begin{vmatrix} f = 2\sqrt{x} & g' = \dfrac{\sin\sqrt{x}}{2\sqrt{x}} \\ f' = \dfrac{1}{\sqrt{x}} & g = -\cos(\sqrt{x}) \end{vmatrix} =$$

$$2x\sin\sqrt{x} - 2\left(-2\sqrt{x}\cos\sqrt{x} - \int \frac{-\cos\sqrt{x}}{\sqrt{x}}\,dx\right) =$$

$$= 2x\sin\sqrt{x} - 2\left(-2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x}\right) + C =$$

$$= 2x\sin\sqrt{x} + 4\sqrt{x}\cos\sqrt{x} - 4\sin\sqrt{x} + C.\ \ MOŻNA\quad INACZEJ$$

$$o)\ \int x^3(\ln x)^2\,dx = \begin{vmatrix} f = (\ln x)^2 & g' = x^3 \\ f' = 2\dfrac{1}{x}\ln x & g = \dfrac{1}{4}x^4 \end{vmatrix} = \frac{1}{4}x^4(\ln x)^2 - \frac{1}{2}\int x^3\ln x\ dx =$$

$$= \frac{1}{4}x^4(\ln x)^2 - \frac{1}{2} = \begin{vmatrix} f = \ln x & g' = x^3 \\ f' = \dfrac{1}{x} & g = \dfrac{1}{4}x^4 \end{vmatrix} = \frac{1}{4}x^4(\ln x)^2 - \frac{1}{2}\left(\frac{1}{4}x^4\ln x - \frac{1}{4}\int x^3\ dx\right) =$$

$$= \frac{1}{4}x^4(\ln x)^2 - \frac{1}{2}\left(\frac{1}{4}x^4\ln x - \frac{1}{16}x^4\right) = \frac{1}{4}x^4(\ln x)^2 - \frac{1}{8}x^4\ln x + \frac{1}{32}x^4 + C$$

$$p^*)\ \int(\sin x)^2\ dx = \begin{vmatrix} f = \sin x & g' = \sin x \\ f' = \cos x & g = -\cos x \end{vmatrix} = -\sin x\cos x + \int(\cos x)^2\ dx =$$

$$= -\sin x\cos x + \int(1 - (\sin x)^2)\ dx = -\sin x\cos x + x - \int(\sin x)^2\ dx$$

w ten sposób otrzymaliśmy równanie na obliczaną całkę $\int(\sin x)^2\ dx$:

$$\int(\sin x)^2\ dx = -\sin x\cos x + x - \int(\sin x)^2\ dx$$

$$2\int(\sin x)^2\ dx = -\sin x\cos x + x \ \Rightarrow\ \int(\sin x)^2\ dx = \frac{-\sin x\cos x + x}{2} + C.$$

$q^*)\quad podobnie\ jak\quad p^*).$

$$r)\ \int e^x\sin x\ dx = \begin{vmatrix} f = \sin x & g' = e^x \\ f' = \cos x & g = e^x \end{vmatrix} = e^x\sin x - \int e^x\cos x\ dx =$$

$$= e^x\sin x - \begin{vmatrix} f = \cos x & g' = e^x \\ f' = -\sin x & g = e^x \end{vmatrix} = e^x\sin x - e^x\cos x - \int e^x\sin x\,dx$$

otrzymujemy równanie:

$$\int e^x\sin x\ dx = e^x\sin x - e^x\cos x - \int e^x\sin x\ dx$$

$$2\int e^x \sin x\ dx = e^x \sin x - e^x \cos x \quad \Rightarrow \quad \int e^x \sin x\ dx = \frac{e^x \sin x - e^x \cos x}{2} + C.$$

s^*) $\int 3^x \cos x\ dx = \begin{vmatrix} f = \cos x & g' = 3^x \\ f' = -\sin x & g = 3^x/\ln 3 \end{vmatrix} = \frac{3^x}{\ln 3}\cos x + \int \frac{3^x}{\ln 3}\sin x\ dx =$

$$= \frac{3^x}{\ln 3}\cos x + \frac{1}{\ln 3}\int 3^x \sin x\ dx = \frac{3^x}{\ln 3}\cos x + \begin{vmatrix} f = \sin x & g' = 3^x \\ f' = \cos x & g = \frac{3^x}{\ln 3} \end{vmatrix} =$$

$$= \frac{3^x}{\ln 3}\cos x + \frac{3^x}{\ln^2 3}\sin x - \frac{1}{\ln^2 3}\int 3^x \cos x\ dx$$

otrzymujemy równanie na całkę $\int 3^x \cos x\ dx$:

$$\int 3^x \cos x\ dx = \frac{3^x}{\ln 3}\cos x + \frac{3^x}{\ln^2 3}\sin x - \frac{1}{\ln^2 3}\int 3^x \cos x\ dx$$

$$\int 3^x \cos x\ dx + \frac{1}{\ln^2 3}\int 3^x \cos x\ dx = \frac{3^x}{\ln 3}\cos x + \frac{3^x}{\ln^2 3}\sin x$$

$$\left(1 + \frac{1}{\ln^2 3}\right)\cdot \int 3^x \cos x\ dx = \frac{3^x}{\ln 3}\cos x + \frac{3^x}{\ln^2 3}\sin x$$

$$\int 3^x \cos x\ dx = \frac{3^x}{1 + \ln^2 3}\left(\ln 3 \cos x + \sin x\right) + C$$

5. Oblicz całkując przez podstawianie (pochodną oznaczmy również przez: $\boldsymbol{f'(x) = \frac{df}{dx}}$) **.**

Całkowanie przez podstawianie

$$\int f\big(t(x)\big)t'(x)dx = \begin{vmatrix} t = \cdots \\ \frac{dt}{dx} = t'(x) \\ dx = \cdots dt \end{vmatrix} = \int f(t)dt = \cdots$$

1. $\int \sqrt{3x-4}\,dx = \begin{vmatrix} t = 3x-4 \\ \frac{dt}{dx} = 3 \\ dx = \frac{dt}{3} \end{vmatrix} = \int \sqrt{t}\,\frac{dt}{3} = \frac{1}{3}\int t^{\frac{1}{2}}\,dt = \frac{1}{3}\cdot\frac{2}{3}\cdot t^{\frac{3}{2}} + C =$

$$= \frac{2}{9}\cdot(3x-4)^{\frac{3}{2}} + C$$

$$4.\int x^2\,(x^3+2)^{34}\,dx=\left|\begin{matrix}t=x^3+2\\ \frac{dt}{dx}=3x^2\\ dx=\frac{dt}{3x^2}\end{matrix}\right|=\int x^2\,t^{34}\,\frac{1}{3x^2}dt=\frac{1}{3}\int t^{34}\,dt=\frac{1}{3}\cdot\frac{1}{35}t^{35}+C=$$

$$=\frac{1}{105}(x^3+2)^{35}+C$$

$$5.\int x\sqrt{x-3}\,dx=\left|\begin{matrix}t=x-3\\ \frac{dt}{dx}=1\\ dx=dt\end{matrix}\right|=\int(t+3)\sqrt{t}\,dt=\int\left(t^{\frac{3}{2}}+3t^{\frac{1}{2}}\right)dt=\frac{2}{5}t^{\frac{5}{2}}+2t^{\frac{3}{2}}+C=$$

$$=\frac{2}{5}(x-3)^{\frac{5}{2}}+2(x-3)^{\frac{3}{2}}+C$$

$$6.\int(x+1)\cos(x^2+2x+1)\,dx=\left|\begin{matrix}t=x^2+2x+1\\ \frac{dt}{dx}=2x+2\\ dx=\frac{dt}{2x+2}\end{matrix}\right|=\int(x+1)\cos t\,\frac{dt}{2(x+1)}=$$

$$=\frac{1}{2}\int\cos t\,dt=\frac{1}{2}\sin t+C=\frac{1}{2}\sin(x^2+2x+1)+C$$

$$7.\int\frac{x+2}{x^2+4x+3}dx=\left\{\begin{matrix}t=x^2+4x+3\\ \frac{dt}{dx}=2x+4\\ dx=\frac{dt}{2(x+2)}\end{matrix}\right\}=\int\frac{x+2}{t}\frac{dt}{2(x+2)}=\frac{1}{2}\int\frac{1}{t}dt=\frac{1}{2}\ln|t|+C$$

$$=\frac{1}{2}\ln|x^2+4x+3|+C$$

$$8.\int xe^{3x^2+1}\,dx=\left\{\begin{matrix}t=3x^2+1\\ \frac{dt}{dx}=6x\\ dx=\frac{dt}{6x}\end{matrix}\right\}=\int xe^t\frac{dt}{6x}=\int e^t\frac{dt}{6}=\frac{1}{6}e^t+C=\frac{1}{6}e^{3x^2+1}+C$$

$$9.\int\frac{\ln x}{x}dx=\left\{\begin{matrix}t=\ln x\\ \frac{dt}{dx}=\frac{1}{x}\\ dx=xdt\end{matrix}\right\}=\int\frac{t}{x}xdt=\int t\,dt=\frac{1}{2}t^2+C=\frac{1}{2}(\ln x)^2+C$$

$$10.\ \int \frac{2\sin x}{4-3\cos x}\,dx = \left\{\begin{array}{l} t = 4-3\cos x \\ \dfrac{dt}{dx} = 3\sin x \\ dx = \dfrac{dt}{3\sin x} \end{array}\right\} = \int \frac{2\sin x}{t}\frac{dt}{3\sin x} = \frac{2}{3}\int \frac{1}{t}\,dt = \frac{2}{3}\ln|t| + C =$$

$$= \frac{2}{3}\ln|4-3\cos x| + C$$

$$11.\ \int \frac{e^x}{e^{2x}+1}\,dx = \left\{\begin{array}{l} t = e^x \\ \dfrac{dt}{dx} = e^x \\ dx = \dfrac{dt}{e^x} \end{array}\right\} = \int \frac{e^x}{t^2+1}\,\frac{dt}{e^x} = \int \frac{1}{t^2+1}\,dt = \operatorname{arctg} t + C =$$

$$= \operatorname{arctg}(e^x) + C$$

$$15.\ \int \frac{\sin x}{\sqrt[3]{\cos x}}\,dx = \left\{\begin{array}{l} t = \cos x \\ \dfrac{dt}{dx} = -\sin x \\ dx = \dfrac{dt}{-\sin x} \end{array}\right\} = \int \frac{\sin x}{\sqrt[3]{t}}\frac{dt}{-\sin x} = -\int \frac{1}{\sqrt[3]{t}}\,dt = -\int t^{-\frac{1}{3}}\,dt$$

$$= -\frac{3}{2}t^{\frac{2}{3}} + C = -\frac{3}{2}(\cos x)^{\frac{2}{3}} + C$$

$$16.\ \int \frac{dx}{\sqrt{2-3x^2}} = \int \frac{dx}{\sqrt{2\left(1-\frac{3}{2}x^2\right)}} = \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\left(1-\frac{3}{2}x^2\right)}} = \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\left(1-\left(\sqrt{\frac{3}{2}}x\right)^2\right)}} =$$

$$= \frac{1}{\sqrt{2}}\int \frac{dx}{\sqrt{\left(1-\left(\sqrt{\frac{3}{2}}x\right)^2\right)}} = \frac{1}{\sqrt{2}}\left|\begin{array}{l} t = \dfrac{\sqrt{3}}{\sqrt{2}}x \\ \dfrac{dt}{dx} = \sqrt{\dfrac{3}{2}} \\ dx = \sqrt{\dfrac{2}{3}}dt \end{array}\right| = \frac{1}{\sqrt{2}}\sqrt{\frac{2}{3}}\int \frac{dt}{\sqrt{(1-t^2)}} = \frac{1}{\sqrt{3}}\arcsin t + C$$

$$= \frac{1}{\sqrt{3}}\arcsin\left(\sqrt{\frac{3}{2}}x\right) + C$$

$$21.\ \int \frac{e^{2x}}{e^{4x}+1}\,dx, \text{wsk.}: t = e^{2x}$$

$$22.\int \frac{1+2x}{x^2+1}\ dx = \int \frac{2x}{x^2+1}\ dx + \int \frac{1}{x^2+1}\ dx = \left| \begin{matrix} t = x^2+1 \\ \frac{dt}{dx} = 2x \\ \frac{dx}{2x} = dt \end{matrix} \right| + \int \frac{1}{x^2+1}\ dx =$$

$$= \int \frac{1}{t}\ dt + \int \frac{1}{x^2+1}\ dx = \ln|t| + \operatorname{arctg} x + C = \ln(x^2+1) + \operatorname{arctg} x + C.$$

$$23.\ \int \frac{\ln^3 x - 1}{x}\ dx = \left| \begin{matrix} t = \ln x \\ \frac{dt}{dx} = \frac{1}{x} \\ dx = x\ dt \end{matrix} \right| = \int t^3 - 1\ dt = \frac{t^4}{4} - t + C = \frac{\ln^4 x}{4} - ln\ x + C.$$

6*. Metody mieszane

$$8.\ \int x \operatorname{tg}^2 x\ dx = |\ można\ zauważyć, że\ \operatorname{tg}^2 x = \frac{\sin^2 x}{\cos^2 x} = \frac{1-\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} - 1\ | =$$

$$= \int x \left(\frac{1}{\cos^2 x} - 1 \right) dx = \int x \frac{1}{\cos^2 x}\ dx - \int x\ dx$$

Obliczamy całkę

$$\int x \frac{1}{\cos^2 x}\ dx = \left| \begin{matrix} f = x & g' = \frac{1}{\cos^2 x} \\ f' = 1 & g = \operatorname{tg} x \end{matrix} \right| = x \operatorname{tg} x - \int \operatorname{tg} x\ dx = x \operatorname{tg} x - \int \frac{\sin x}{\cos x}\ dx.$$

Zauważmy, że

$$\int \frac{\sin x}{\cos x}\ dx = \left| \begin{matrix} t = \cos x \\ dt = -\sin x\ dx \end{matrix} \right| = -\int \frac{1}{t} dt\ = -\int \frac{1}{t} dt = -\ln|\cos x| + C,$$

zatem

$$x \operatorname{tg} x - \int \frac{\sin x}{\cos x}\ dx = x \operatorname{tg} x + \ln|\cos x| + C.$$

W ostateczności:

$$\int x \operatorname{tg}^2 x\ dx = \int x \frac{1}{\cos^2 x}\ dx - \int x\ dx = x \operatorname{tg} x + \ln|\cos x| - \frac{x^2}{2} + C.$$

$$9.\int \sqrt{x} \cdot \cos(\sqrt{x})\ dx = \left| \begin{matrix} t = \sqrt{x} \Rightarrow x = t^2 \\ \frac{dt}{dx} = \frac{1}{2\sqrt{x}} \\ dx = 2\sqrt{x}\ dt \Rightarrow dx = e^t dt \end{matrix} \right| = \int \sqrt{x} \cdot \cos(t)\ 2\sqrt{x}\ dt =$$

$= 2\int x \cdot \cos(t)\ dt = 2\int t^2 \cdot \cos(t)\ dt\ =$ | teraz dwa razy przez części| $= \cdots$

10. $\int \frac{dx}{\sqrt{3+2e^x}} = \left| \begin{matrix} t = \sqrt{3+2e^x} \Rightarrow & e^x = \frac{1}{2}(t^2-3) \\ dt = \frac{2e^x}{2\sqrt{3+2e^x}}dx \Rightarrow & dx = \frac{2t}{t^2-3}dt \end{matrix} \right| = \int \frac{1}{t}\ \frac{2t}{t^2-3}dt$

$$= 2\int \frac{1}{t^2-3}dt = \frac{1}{\sqrt{3}}\int \frac{1}{t-\sqrt{3}} - \frac{1}{t+\sqrt{3}}\ dt$$

$$= \frac{1}{\sqrt{3}}\left(\ln\left|\sqrt{3+2e^x}-3\right| - \ln\left|\sqrt{3+2e^x}+3\right|\right).$$